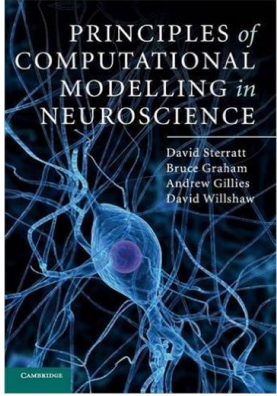
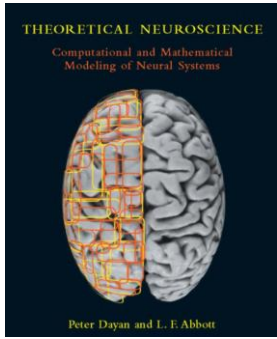
Miscellaneous


PRINCIPLES of
COMPUTATIONAL
MODELLING in
NEUROSCIENCE
David Sterratt
Bruce Graham
Andrew Gillies
David Willshaw
CAMBRIDGE

In this course we will need to program in Python and NEURON. In order to get the same errors ;) we should use Google Colab. Please ensure that you have a gmail account in order to use Google Colab.

This course is based mainly on the books: “Principles of Computational Modelling in Neuroscience” by David Sterratt, Bruce Graham, Andrew Gillies, David Willshaw, Cambridge University Press, and “Theoretical neuroscience” by Peter Dyan, L.F. Abbott, MIT Press.


THEORETICAL NEUROSCIENCE
Computational and Mathematical
Modeling of Neural Systems
Peter Dayan and L. F. Abbott

The format will be: lecture followed by practical implementations. Nevertheless, feel free to ask for recaps/more exercises or to elaborate more in something you might find interesting and want further details on.

NO final written examination in course material. Grade will be based on a project followed by a 10min presentation. During the course we will suggest ~4 projects that you could choose from, but feel free also to suggest one of your interest!

Syllabus

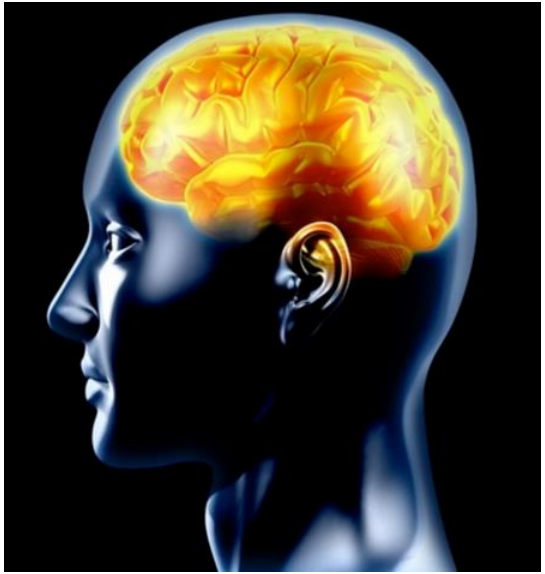
- 08/03/2022 (12:00-14:00) **Lecture 1:** Introduction to computational modeling in Neuroscience
- 10/03/2022 (09:00-12:00) Practical tutorial 1
- 15/03/2022 (09:00-11:00) **Lecture 2:** Spiking neuronal models
- 17/03/2022 (09:00-12:00) Practical tutorial 2
- 22/03/2022 (12:00-15:00) Practical tutorial 3
- 24/03/2022 (10:00-12:00) **Lecture 3:** Connections between neurons (Synapses)
- 29/03/2022 (09:00-11:00) **Lecture 4:** Hodgkin-Huxley equations and biophysical neuronal models
- 31/03/2022 (09:00-12:00) Practical tutorial 4
- 05/04/2022 (13:00-15:00) **Lecture 5:** Synaptic Plasticity (LTP, LTD)
- 12/04/2022 (13:00-16:00) Practical tutorial 5 and Recap
- 13/05/2022 (23:59) Final project delivery
- 17/05/2022 (09:00-11:00) Project presentations

Computational Modelling

Theories are possible explanations for how the phenomena under investigation arise.

We call a theory that has been specified by sets of equations a **mathematical model**.

A particular mathematical model may have an **analytical solution** that allows exact calculation of quantities, or may require a **numerical solution** (referred to as **computational modelling**).



Scientific effort to understand how the brain works

Image adopted: FENS 2016

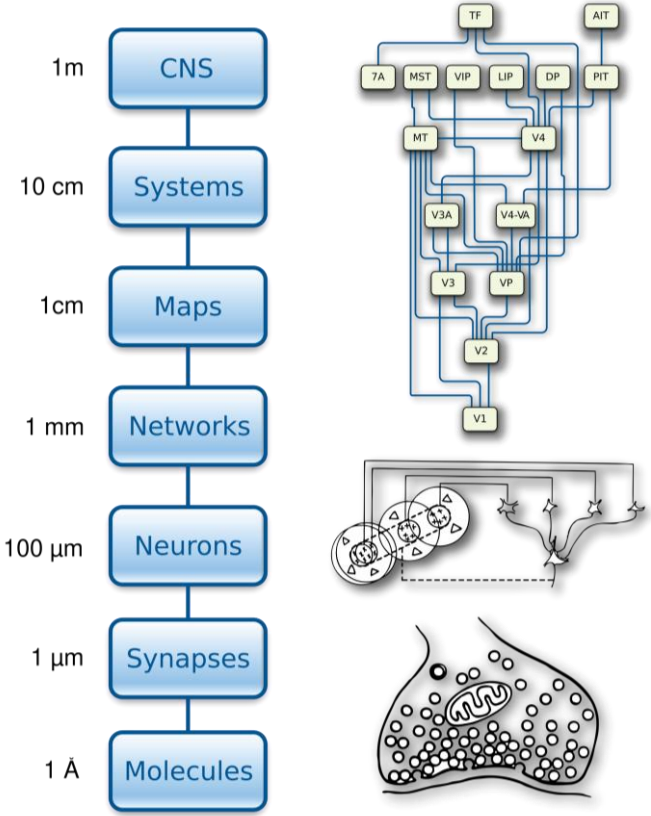
Why Model?

1. **Explain** experimental results – Gain insights into key biological parameters.

2. **Suggest** possible computational (functional) role for the modeled system (**predictions**).

3. **Inspire** implementations in other fields.

Levels of analysis

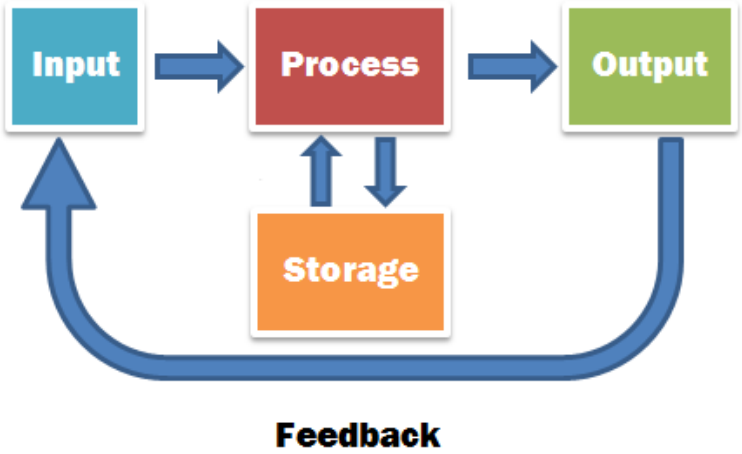


Source: cnl.salk.edu

Source: britannica.com

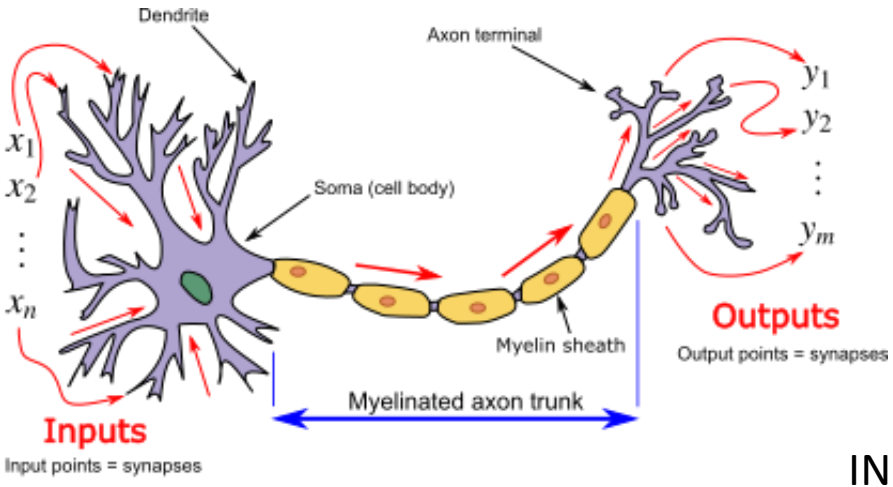
Source: neurosciencenews.com

Neurons/neuronal circuits implement input-output relations

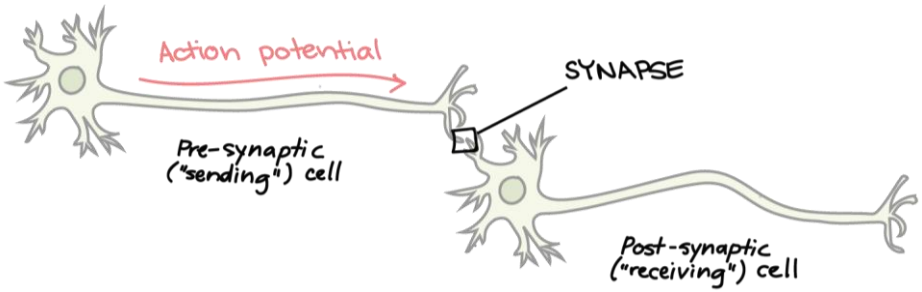


Source: blendspace.com

Levels of details

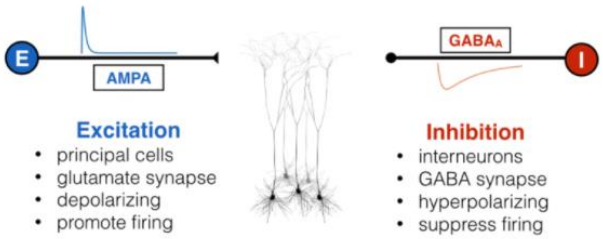


Source: wikipedia.org



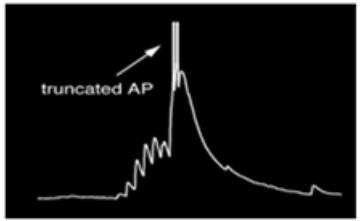
Source: khanacademy.org

INPUTS



Source: voyteklab.com

OUTPUTS



Source: coursera.org/learn/synapses

Which level of detail?

The level of description (variable of interest) can range from:

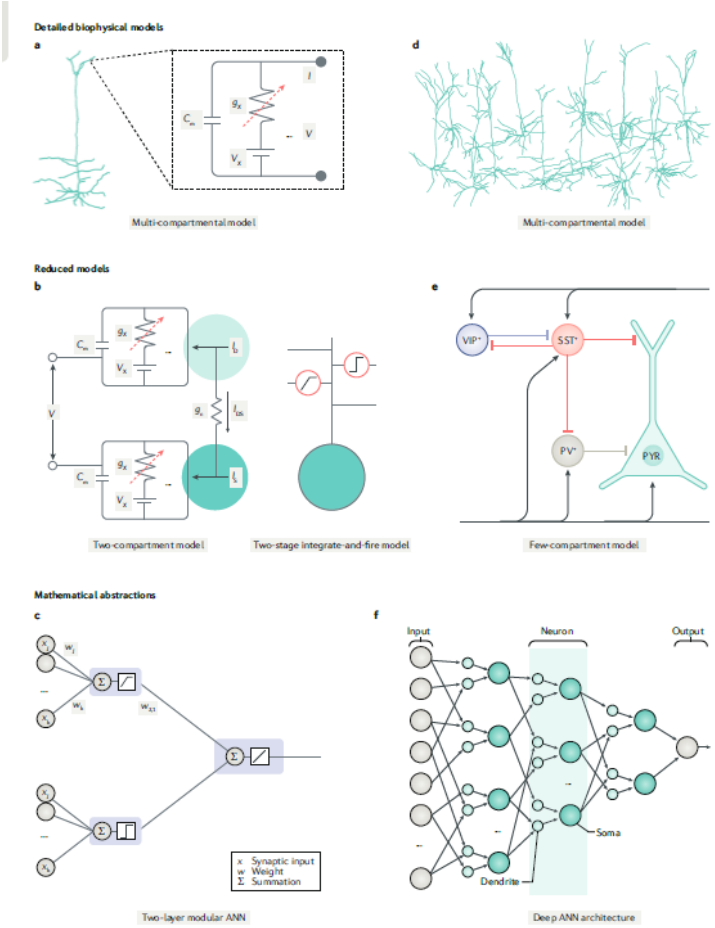
- the average firing rate of a neuron (rate models).
- the voltage of a neuron (spiking models).
- The biological realism (biophysics/geometry) of the voltage of the neurons (compartmental models).

Which level of detail?

Models that are constructed at the same level of analysis may be constructed to different levels of detail.

Which level of detail?
Depends on the question(s) asked.

Occam's razor: the best model is the one detailed enough to address the question asked but requiring the smallest possible number of parameters/ assumptions.



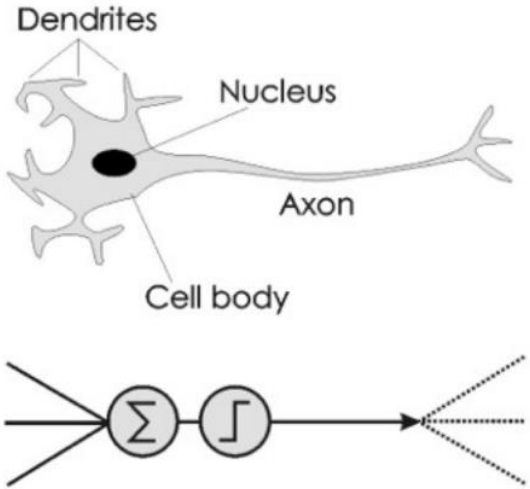
Poirazi, Papoutsi, Nat. Rev. Neurosci. 2020

Rate-based models

McCulloch-Pitts (idea)

McCulloch and Pitts (1943): pioneers to formally define neurons as computational elements

The idea: explore simplified neural models to get the essence of neural processing by ignoring irrelevant detail and focusing in what is needed to do a computational task

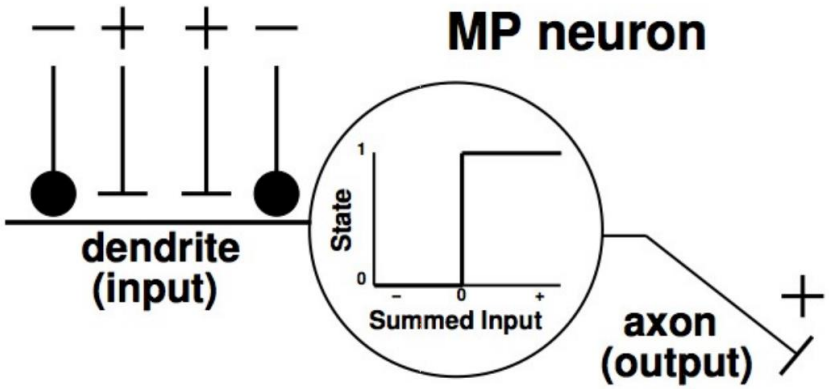


McCulloch-Pitts (idea)

- McCulloch and Pitts knew that spikes (action potential) somehow carry information through the brain.
- Each spike would represent a binary 1, each lack of spike would represent a binary 0.
- They showed how spikes could be combined to do logical and arithmetical operations.
- From modern perspective there is a design problem for these circuit elements: brevity of a spike (1 ms) compared to the interval between spikes (>50 ms) implies a duty cycle of < 2%.

McCulloch-Pitts (rule)

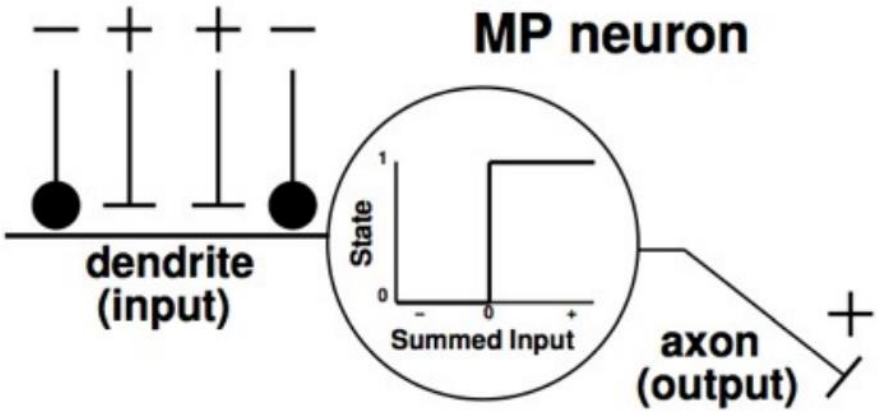
MP neurons are binary: take as input and produce as output only 0's or 1's

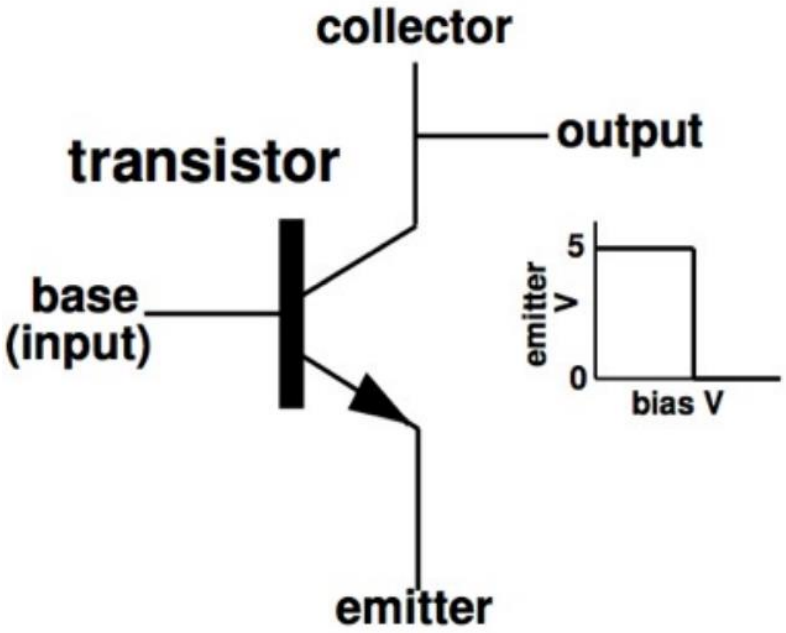


Rule: activations from other neurons (inputs) are summed at the neuron soma and outputs 1 if threshold is reached and 0 if not.

McCulloch-Pitts (analogy)

MP neurons function much like a transistor:





In artificial neural networks, inputs come from the outputs of other MP neurons (as transistors in a circuit)

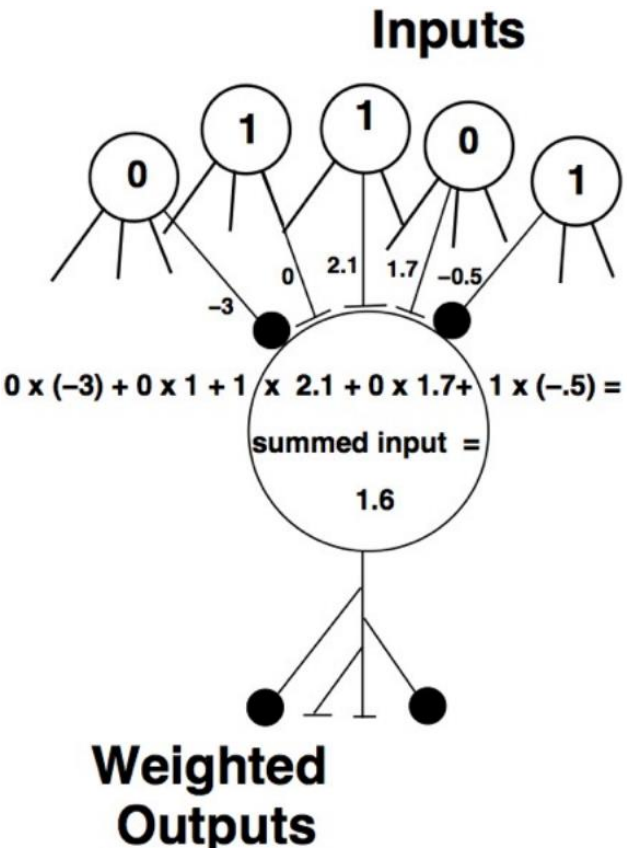
McCulloch-Pitts (nomenclature)

State: the degree of activation of a single neuron

Weight: the strength of the connection between two neurons

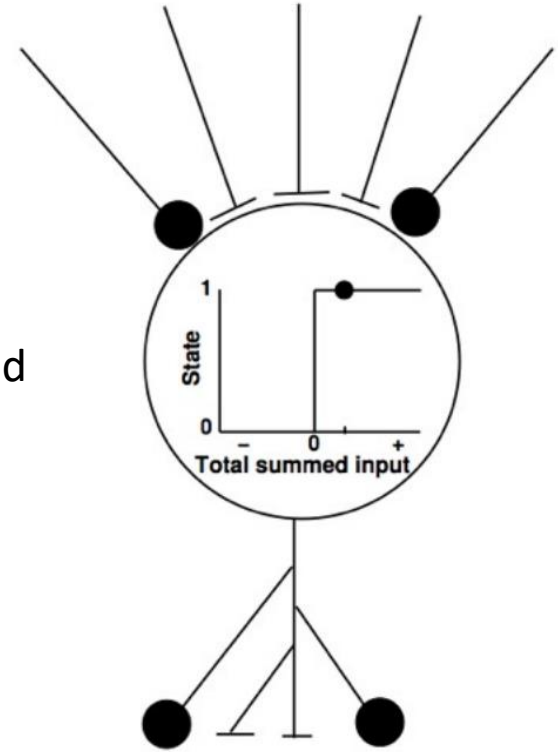
Bias: determines when a neuron will be fired

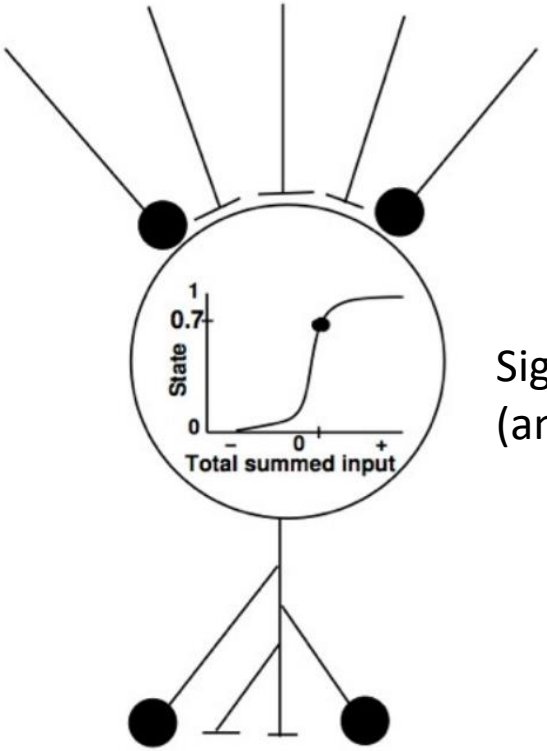
Update rule: determines how input to a neuron is translated into the state of that neuron



McCulloch-Pitts (states)

Sharp threshold
(digital)





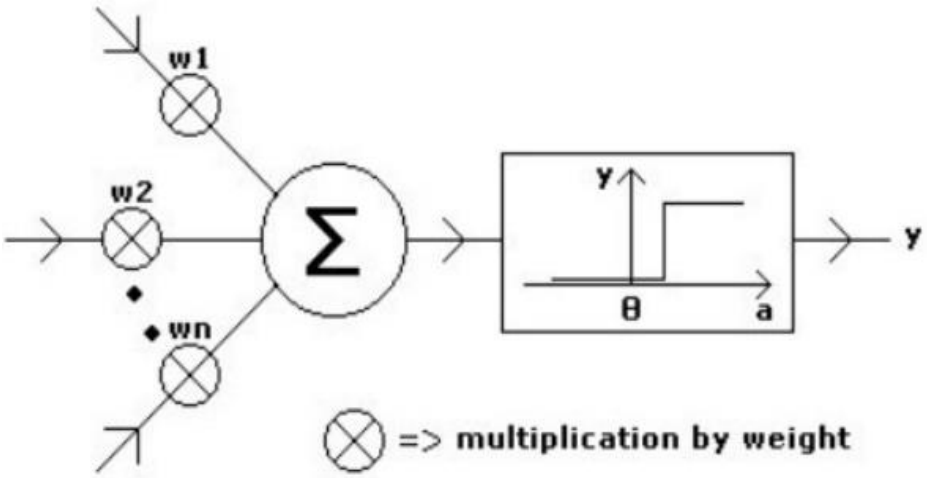
Sigmoid function
(analog)

McCulloch-Pitts (states)

In mathematical terms:

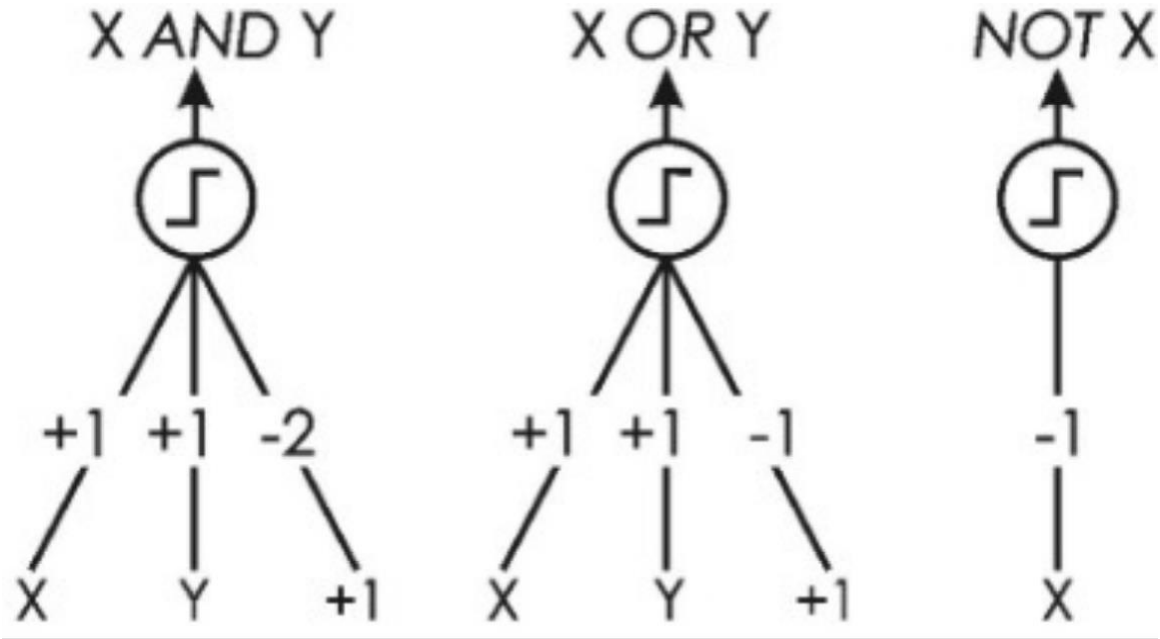
$$y = \phi\left(\sum_{i=1}^{n} w_i \cdot x_i + b\right)$$

where $\boldsymbol{\phi}$ represents a *threshold* or a *sigmoid* function

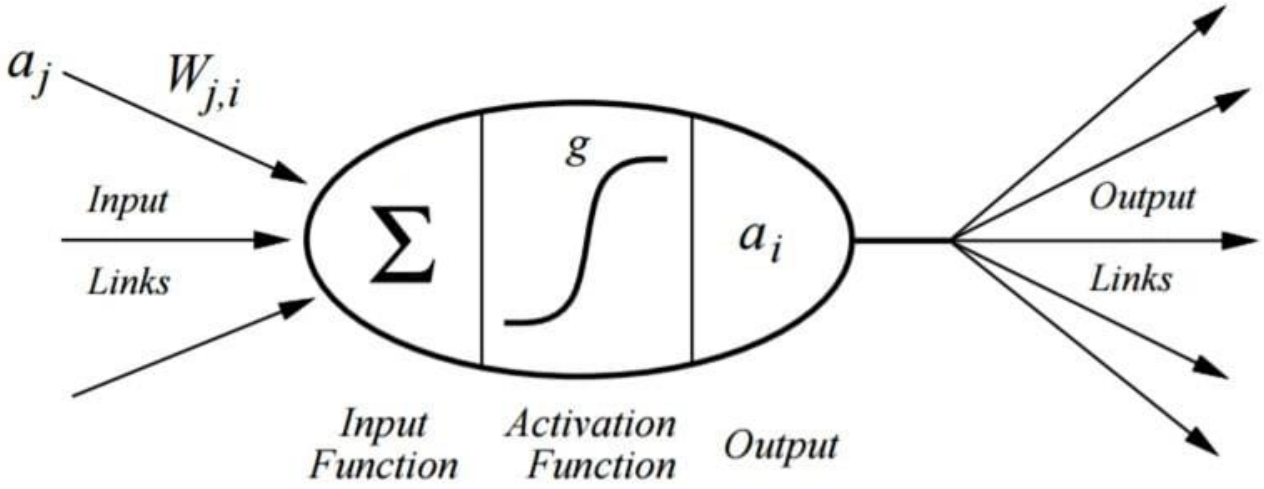


McCulloch-Pitts (logic)

MP neurons are capable of logic functions

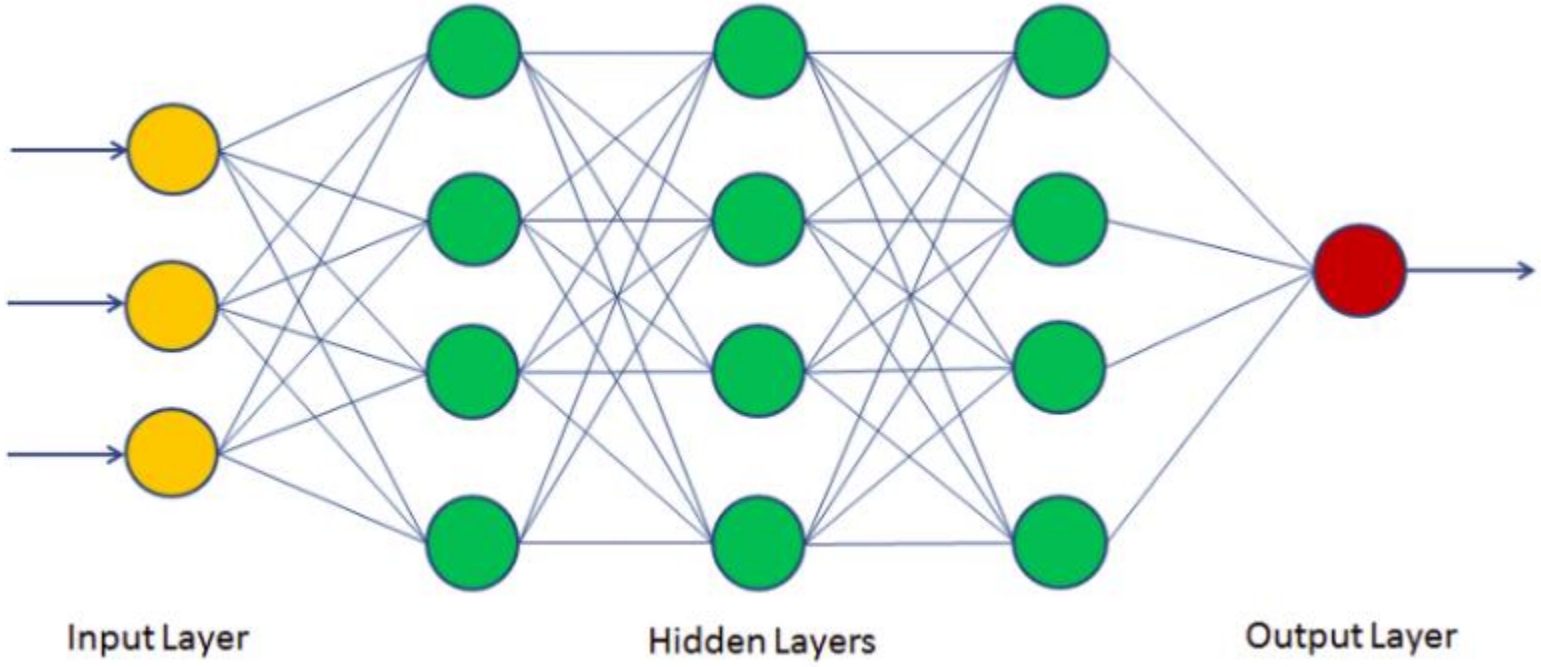


Perceptron



$$a_i = \mathrm{g}\left(\sum_j W_{j,i} a_j + b_i\right)$$

Multilayer perceptrons (ANNs)

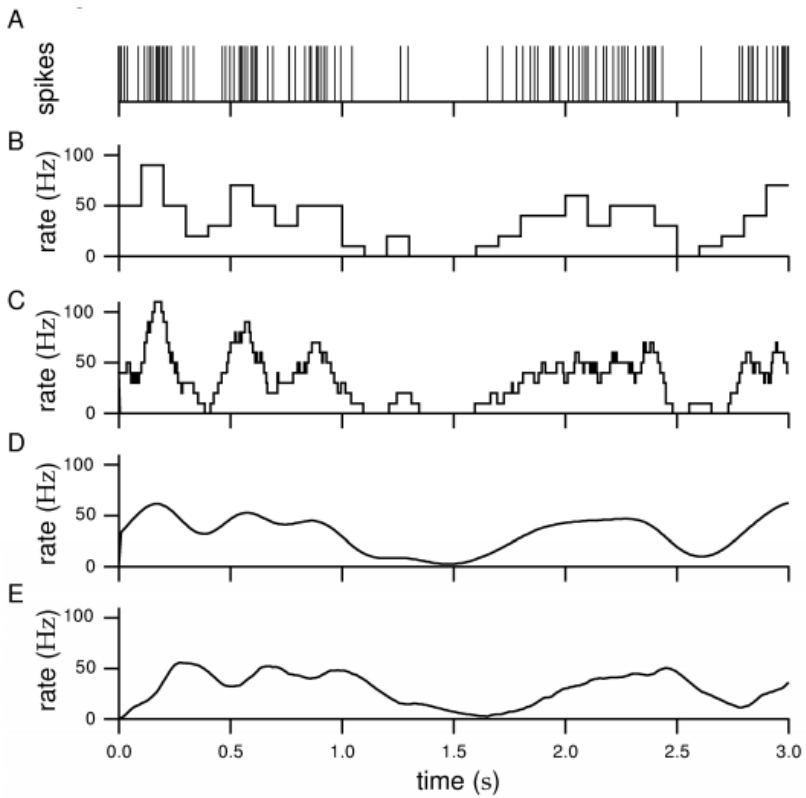


Source: machinelearninggeek.com

How to incorporate time?

- Regardless of the transmission by action potentials the information lies in the average firing rate.

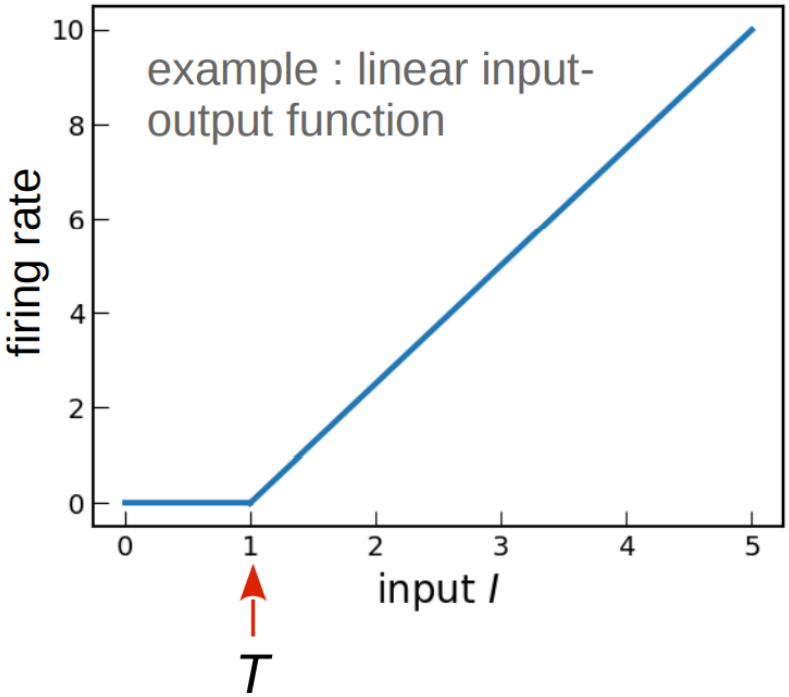
- The rate code assumes an independent-spike code, that is, the generation of each spike is independent of all the other spikes in the network.



Adopted from Dyan, Abbott, Theoretical Neuroscience

Rate-based models

Phenomenological description of the input-output function



$$\tau \frac{d\mathbf{r}}{dt} = -\mathbf{r} + F\left(I_{syn} + I_{ext} - T\right)$$

$\mathbf{r}$: output of the neuron – firing rate

τ : membrane time constant

$F(\cdot)$: input-output transfer function (e.g., sigmoid, threshold)

I_{syn}: synaptic input

I_{ext} : external current

T : firing threshold

Rate-based models

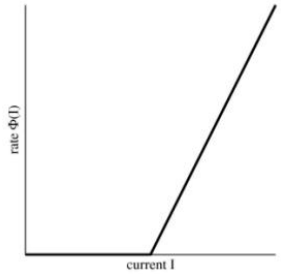
In a rate-based model (also called: firing rate model, neural mass model, neural field model, Wilson-Cowan model), one describes the activity (i.e., instantaneous firing rate) of a population of neurons at a given location by a single analog variable:

$$\tau \dot{r}\left(x,t\right) = -r\left(x,t\right) + F\left(I\left(x,t\right) + \int J\left(\left|x-y\right|\right) r\left(y,t\right) dy\right)$$

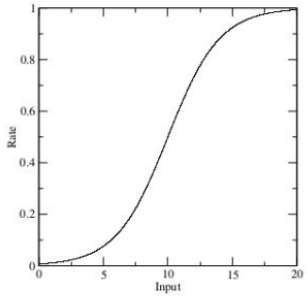
$r(x,t)$: output of the neuron – firing rate of neuron at location x at time t
τ : time constant of firing rate dynamics
$F(\cdot)$: transfer function (f-I curve)
$I(x,t)$: external input
$J(x,y)$: strength of synaptic connections between neurons at location x and y

The transfer function (F *or* Φ)

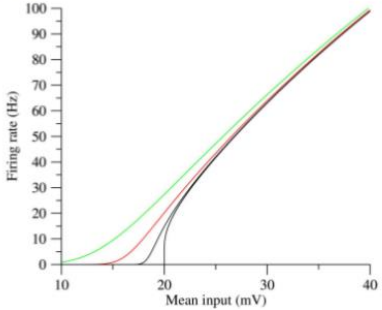
Threshold linear $\Phi(x)=[x-T]_+$



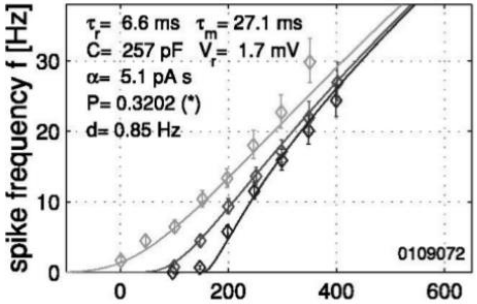
Sigmoidal $\Phi(x)=1/(1+\exp(-\beta(x-T)))$



f-I curve of a specific spiking neuron model



f-I curve of a real neuron [Rauch et al 2003]



Rauch et al., 2003

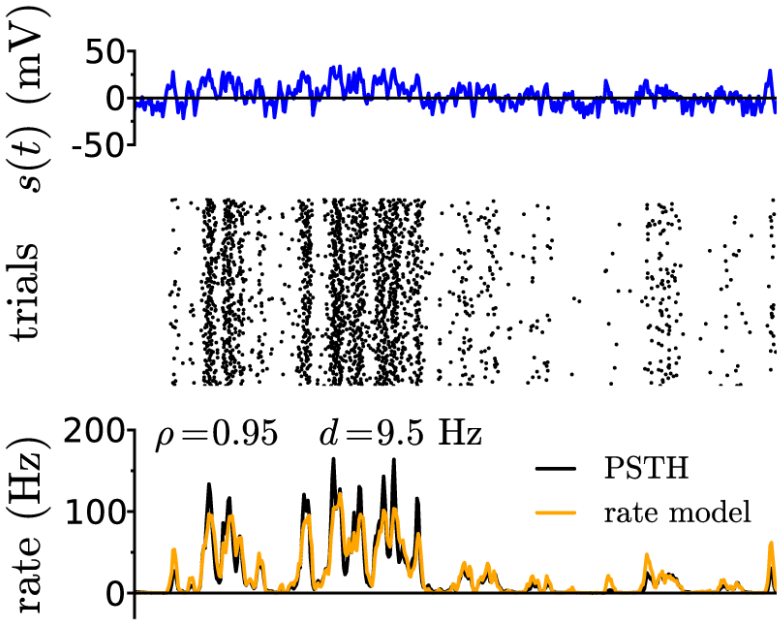
From populations of individual neurons to a rate model

The population activity of homogeneous populations of

- Stochastic binary neurons
- Stochastic spiking neurons

can sometimes be shown to be well approximated by firing rate equations.

$$\tau \frac{d\mathbf{r}}{dt} = -\mathbf{r}(t) + F\big(I_{ext}(t) + J\mathbf{r}(t)\big)$$



Ostojic, Brunel, 2011

*PSTH: Peristimulus time histogram

Rate models of local networks of neurons

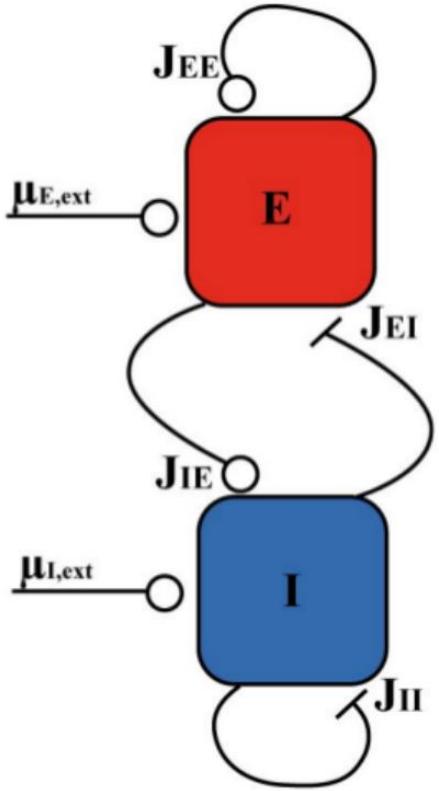
n sub-populations described by their **average firing rate** $r_i, i = 1, \dots, n$

$$\tau \dot{r}_i = -r_i + F_i\left(I_{ext} + \sum_j J_{ji} r_j\right), \dot{r}_i \equiv \frac{dr_i}{dt}$$

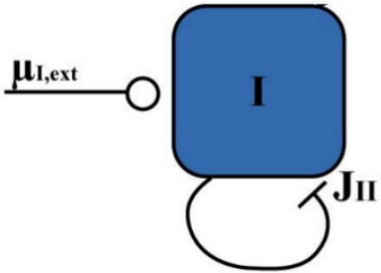
Example: E-I network (Wilson and Cowan, 1972)

$$\tau_E \frac{d\mathbf{r}_E}{dt} = -\mathbf{r}_E + F_E\left(\mu_{ext,E} + J_{EE}\mathbf{r}_E - J_{EI}\mathbf{r}_I\right)$$

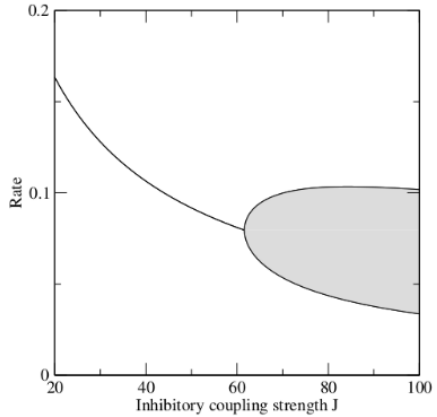
$$\tau_I \frac{d\mathbf{r}_I}{dt} = -\mathbf{r}_I + F_I\left(\mu_{ext,I} + J_{IE}\mathbf{r}_E - J_{II}\mathbf{r}_I\right)$$



Example: Inhibitory network rate model with delays - oscillations

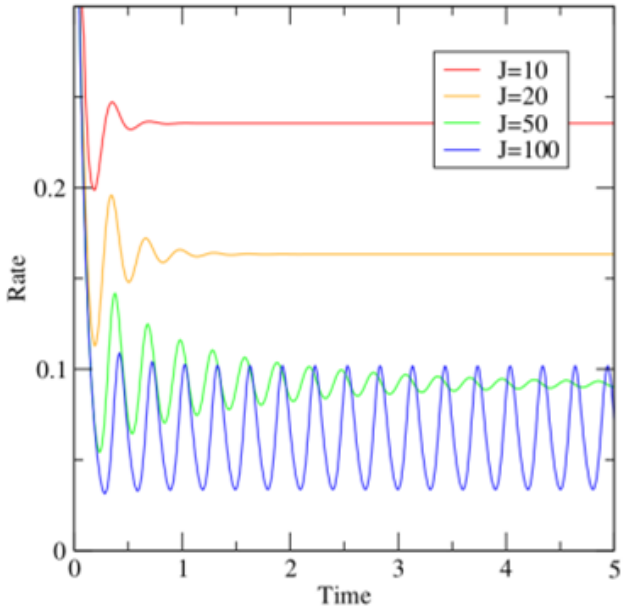


$$\tau \frac{d\mathbf{r}_I}{dt} = -\mathbf{r}_I(t) + F[I_{ext} - J_{II}\mathbf{r}_I(t-D)]$$

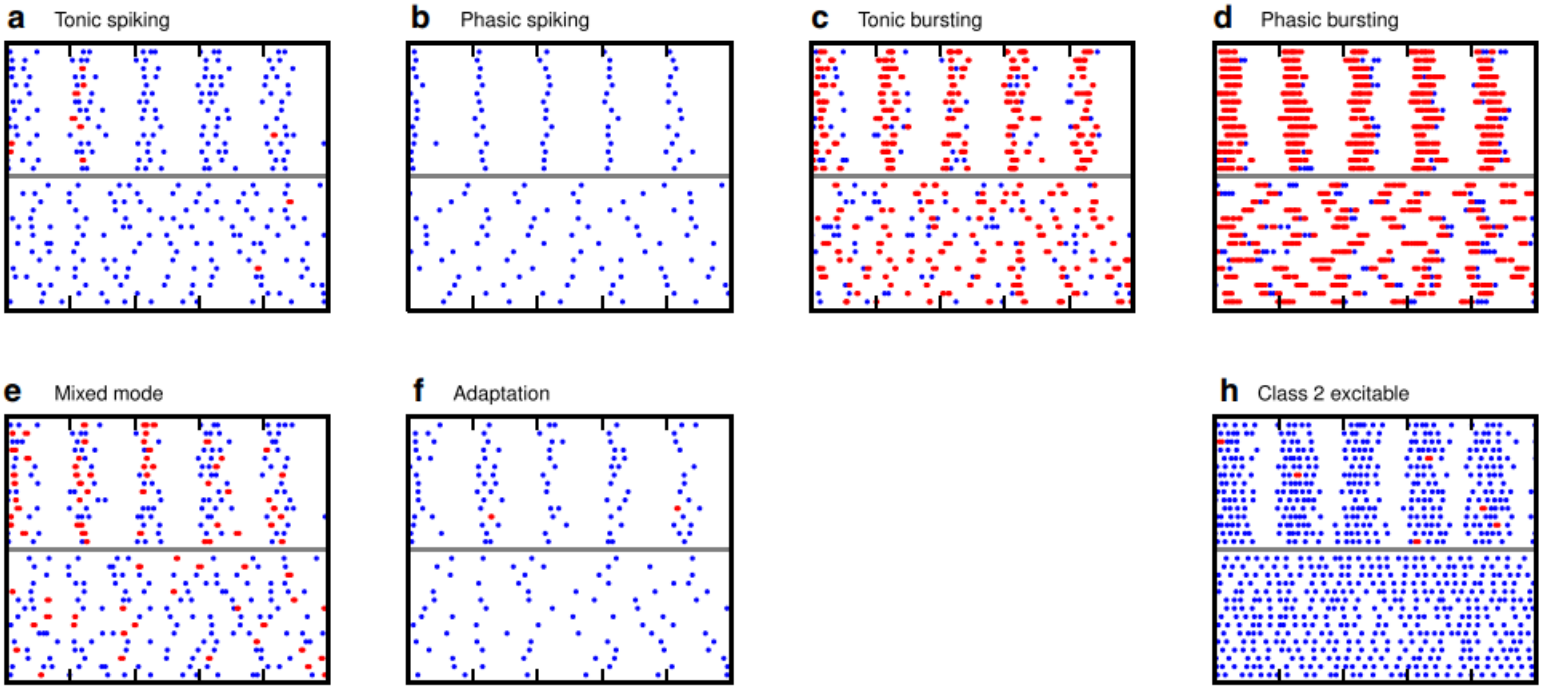


- Oscillations at a frequency f_c appear when $J_{II} > J_c$
- For $D \gg \tau$, $J_c \sim \pi\tau/(2D)$, $f_c \sim 1/(4D)$
- Frequency controlled by synaptic delays
- Fast oscillations in cortex and/or hippocampus

- More info: [Brunel, Wang, 2003]

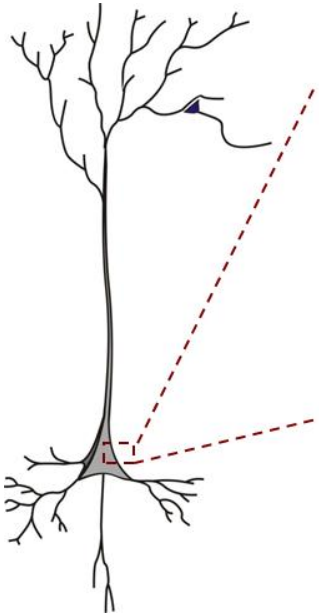


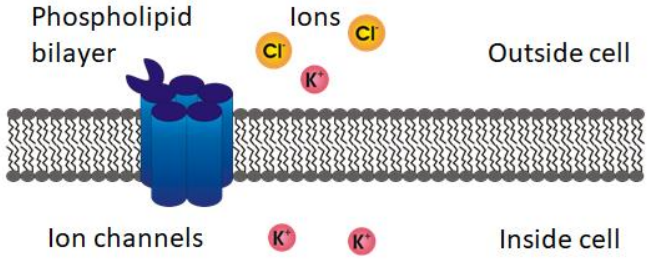
Broad repertoire of behaviors



Heiberg et al., 2018

Spiking models

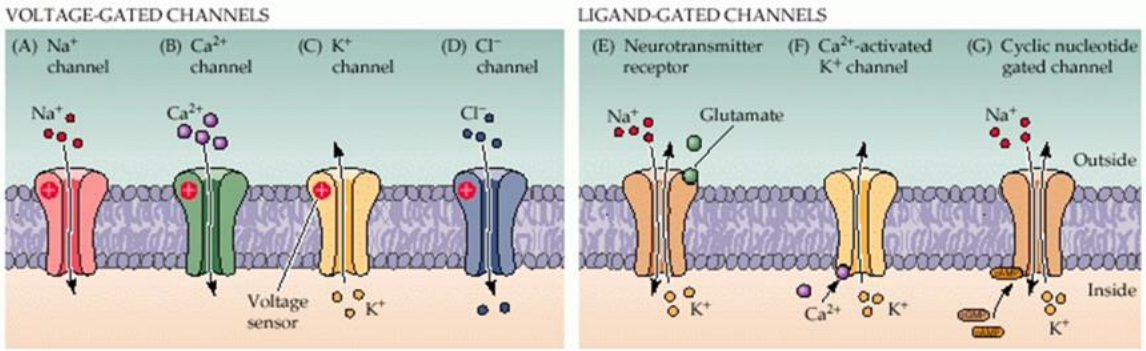




Membrane is charged by currents flowing into it and firing an action potential and discharging when the membrane potential exceeds a threshold.

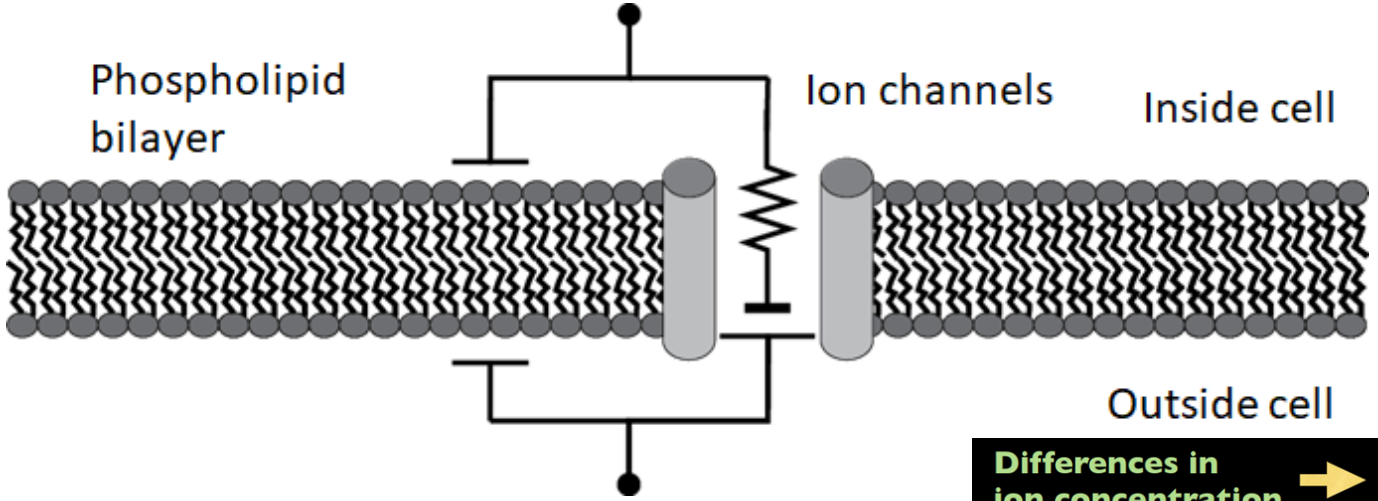
Types of ion channels

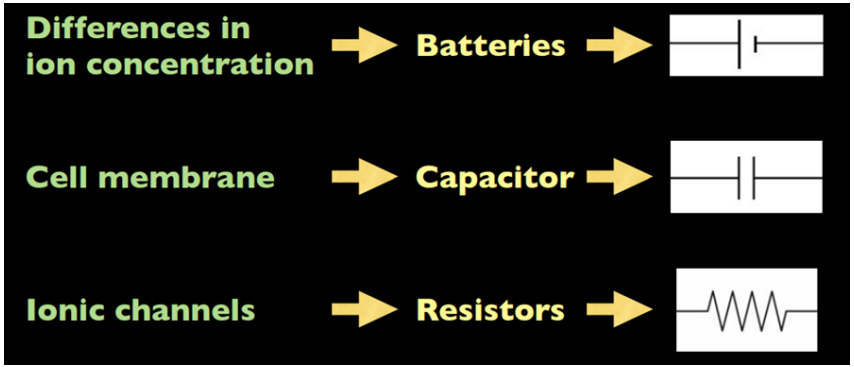
- **Passive** ion channels -- no change to permeability due to external factors
- **Active** ion channels – permeability is dependent upon factors such as:
  - Membrane potential (i.e., voltage dependent)
  - Ion concentrations (e.g., calcium-dependent)
  - Ligands (e.g., neurotransmitters)



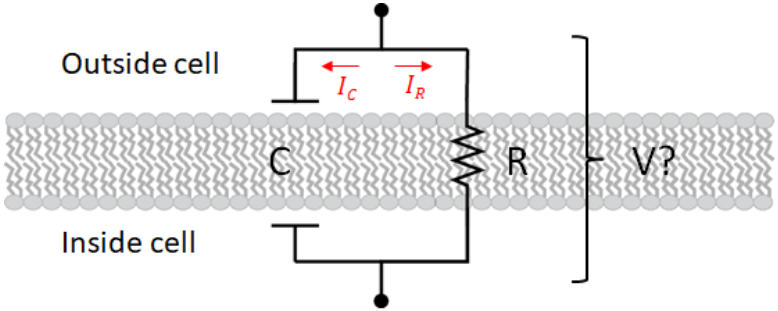
Neuroscience, 2nd edition

Membrane as an RC circuit





Electric circuit of the membrane patch



Kirchoff's current law: $I_{ext} = I_R + I_C$ or $0 = I_R + I_C - I_{ext}$

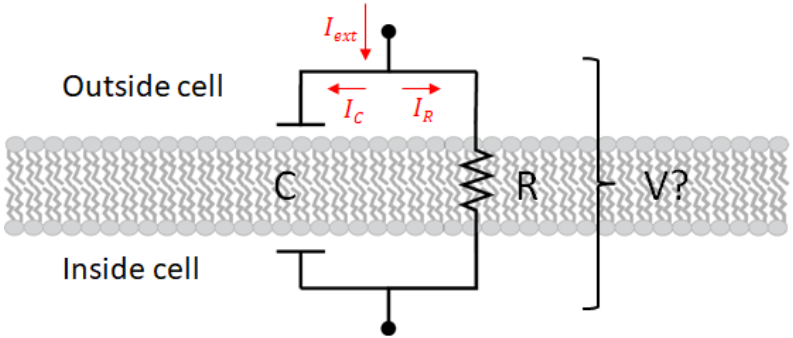
Ohm's law: $V = IR$

Capacitance: $q = CV$ $\frac{dq}{dt} = C\frac{dV}{dt}$

$$I_C = C\frac{dV}{dt}$$

$$\boxed{C\frac{dV}{dt} = -\frac{V}{R} + I_{ext}}$$

Electric circuit of the membrane patch



$$\boxed{C\frac{dV}{dt} = -\frac{V}{R} + I_{ext}}$$

$$RC\frac{dV}{dt} = -V + I_{ext}R$$

$$\tau\frac{dV}{dt} = -V + V_\infty$$

$$\tau = RC$$

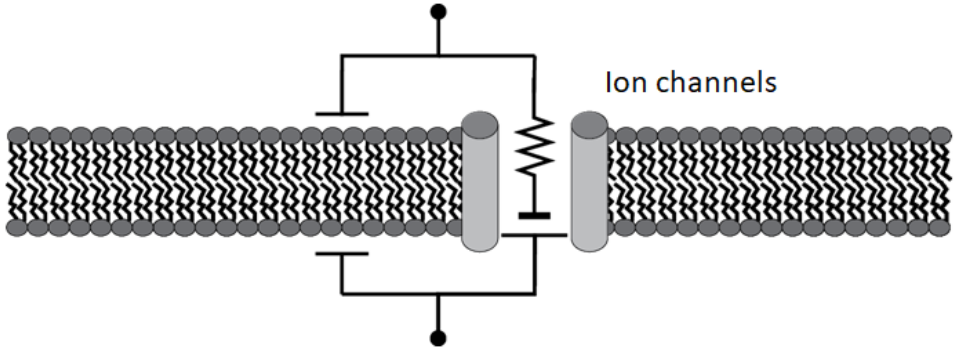
Membrane time constant

$$V_\infty = I_{ext}R$$

Electric circuit of the membrane patch

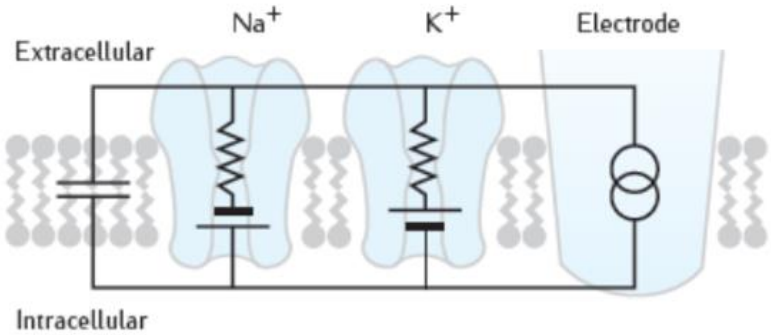
Neurons have a ‘battery’ thanks to transmembrane ion gradients

Outside cell: High Na^+, Cl^-, Ca^{2+}



Inside cell: High K^+

Electric circuit of the membrane patch



Source: Principles of Computational Modelling in Neuroscience

The equivalent electrical circuit of a patch of membrane

$$I_X = g_X(V - E_X)$$

Where X is the ion of interest, E_X its equilibrium potential, and g_X is the conductance.

The term in parenthesis is called **driving force**.

The **Nernst equation** to calculate the **equilibrium (reversal) potential** of a single ion:

$$E_X = \frac{RT}{z_x F} \ln \frac{[X]_{\text{out}}}{[X]_{\text{in}}}$$

R: gas constant, T: temperature, z_X: valence of X, F: Faraday constant, $[X]$: intracellular and extracellular concentration of X.

Typical values: $\mathrm{E_K} \approx -80\ \mathrm{mV},\ \mathrm{E_{Na}} \approx 62\ \mathrm{mV}, \mathrm{E_{Cl}} \approx -65\ \mathrm{mV}$

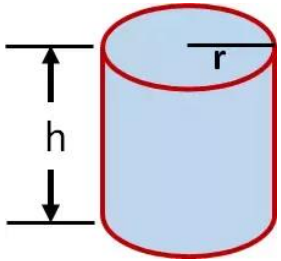
The **Goldman-Hodgkin-Katz equation** to calculate the **equilibrium (reversal) potential** of a membrane:

$$V_{\text{rest}} = \frac{\mathrm{g_K E_K} + \mathrm{g_{Na} E_{Na}} + \mathrm{g_{Cl} E_{Cl}}}{\mathrm{g_K} + \mathrm{g_{Na}} + \mathrm{g_{Cl}}}$$

Note that: $\mathrm{g_K} \gg \mathrm{g_{Na}} + \mathrm{g_{Cl}}$

Specific capacitance and resistance

Both $\mathrm{R_m}$ and $\mathrm{C_m}$ are dependent on surface area of membrane $\boldsymbol{A}$ (open cylinder, $A = 2\pi \mathrm{rh}$, r: radius, h: length).



Therefore we define the size-independent versions of the specific membrane capacitance $\mathrm{C_m}$ (F/cm^2) and specific membrane resistance R_m $(\Omega \cdot \mathrm{cm}^2)$, as $\mathbf{c_m} = \mathbf{C_m}\boldsymbol{A}$ and $\mathbf{r_m} = \mathbf{R_m}/\boldsymbol{A}$, respectively.

Membrane time constant $\boldsymbol{\tau}_\mathbf{m} = \mathbf{R_m C_m} = \mathbf{r_m c_m}$ sets the basic timescale for changes in the membrane potential (is typically between 10 and 100ms).